Math 31 Curriculum

Advanced Algebra | Middlesex School Mathematics | OpenStax: OpenStax Viewer

COURSE AT A GLANCE

Review Content

REVIEW

Primary: 2.2 Use a Problem Solving Strategy; 5.2 Properties of Exponents and Scientific Notation; 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials

UNIT LEARNING OBJECTIVES

I can calculate a percentage, percent increase, or percent decrease.

I can simplify an expression using exponent properties.

I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.

I can expand a product of two binomials.

Linear Functions and Function Notation

REQUIRED

Primary: College Algebra 2e 3.1 Functions and Function Notation; College Algebra 2e 3.3 Rates of Change and Behavior of Graphs; College Algebra 2e 3.2 Domain and Range; College Algebra 2e 4.1 Linear Functions; Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables

UNIT LEARNING OBJECTIVES

I can interpret function notation as a statement about input and output.

I can determine whether a graph or table represents a function.

I can evaluate a function or solve for an input using a graph, table, or formula.

I can identify independent and dependent variables in a relationship.

I can calculate average rate of change from two points.

I can interpret interval notation and identify increasing or decreasing intervals.

I can determine total change from an average rate of change over an interval.

I can determine and interpret the units of an average rate of change.

I can identify a linear function from a table, graph, or formula.

I can interpret the slope and intercept of a linear model.

I can determine the intersection of two linear relationships graphically or algebraically.

I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

Function Behavior and Structure

REQUIRED

Primary: 3.1 Functions and Function Notation; 3.2 Domain and Range; 3.5 Transformation of Functions; 3.4 Composition of Functions; 3.7 Inverse Functions; 3.3 Rates of Change and Behavior of Graphs

UNIT LEARNING OBJECTIVES

I can determine whether a relation is a function.

I can evaluate a function from a formula, graph, or table.

I can determine domain and range from a formula, graph, or table.

I can determine algebraic domain restrictions caused by denominators and even roots.

I can graph a piecewise function from its formula.

I can write a piecewise formula that represents a graph.

I can graph a transformed function using shifts, stretches, compressions, and reflections.

I can describe how a transformation affects domain, range, or asymptotes.

I can evaluate a composition from formulas, graphs, or tables.

I can write a formula for a composition of two functions.

I can find an inverse function algebraically.

I can evaluate and interpret an inverse from a graph, formula, or table.

I can determine and interpret the units of an inverse function.

I can explain how the domain and range of a function relate to those of its inverse.

I can determine the range of a function by analyzing the domain of its inverse.

I can verify algebraically that two functions are inverses.

I can interpret the meaning and units of a composition.

I can classify a graph or table as concave up or concave down using changes in average rate of change.

Function Behavior and Structure (Continued)

REQUIRED

UNIT LEARNING OBJECTIVES - CONTINUED

I can determine whether a function is invertible using the horizontal line test.

I can graph an inverse function as a reflection across the line `y = x`.

Quadratic Functions and Equations

REQUIRED

Primary: 5.1 Quadratic Functions; 2.5 Quadratic Equations
Supplemental: 5.3 Graphs of Polynomial Functions; 5.5 Zeros of Polynomial Functions

UNIT LEARNING OBJECTIVES

I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.

I can connect standard, factored, and vertex forms to features of a quadratic graph.

I can rewrite a quadratic function in a form that reveals a requested feature.

I can solve a quadratic equation by factoring or using the quadratic formula.

I can graph a quadratic function using its algebraic structure and key features.

I can construct a quadratic function from specified zeros, a vertex, or graphical information.

I can create and interpret a quadratic trajectory model.

Exponential Functions and Models

REQUIRED

Primary: 6.1 Exponential Functions; 6.2 Graphs of Exponential Functions

UNIT LEARNING OBJECTIVES

I can distinguish linear change from exponential change using differences, ratios, or percent change.

I can identify initial value and growth or decay factor in an exponential model.

I can write an exponential function from a table, graph, or context.

I can evaluate and interpret an exponential model.

I can calculate growth under periodic compounding.

I can calculate growth under continuous compounding.

I can determine an exponential formula from two points or other sufficient data.

I can determine domain, range, intercepts, and asymptotes of an exponential function.

I can compare the growth rates of two exponential functions.

I can determine the percent growth or decay represented by a continuous exponential model.

Logarithmic Functions and Models

REQUIRED

Primary: 6.3 Logarithmic Functions; 6.5 Logarithmic Properties; 6.6 Exponential and Logarithmic Equations; 6.7 Exponential and Logarithmic Models; 6.4 Graphs of Logarithmic Functions
Supplemental: 6.2 Graphs of Exponential Functions

UNIT LEARNING OBJECTIVES

I can convert between logarithmic and exponential forms.

I can evaluate a logarithm using its exponential meaning.

I can expand or condense an expression using product, quotient, and power properties.

I can solve an exponential equation using logarithms.

I can solve a logarithmic equation using exponential form and check domain restrictions.

I can calculate and interpret doubling time or half-life.

I can determine domain, range, intercepts, and asymptotes of a logarithmic function.

I can explain how exponential and logarithmic functions undo one another.

I can solve a multi-step exponential or logarithmic equation.

I can determine an intersection of exponential functions graphically or algebraically.

I can rewrite a discrete exponential model as an equivalent continuous exponential model.

I can solve and interpret an exponential or logarithmic modeling problem.

Extension Content

EXTENSION

Primary: 3.6 Absolute Value Functions; 3.5 Transformation of Functions; 6.7 Exponential and Logarithmic Models

UNIT LEARNING OBJECTIVES

I can analyze an absolute-value function using transformations.

I can determine even or odd symmetry from a graph or formula.

I can interpret a contextual logarithmic scale.

IN-DEPTH CURRICULUM

SKILL PROGRESSION	I Introduce	R Reinforce	D Deepen	A Apply

Review Content

REVIEW

OpenStax coverage: **Primary:** 2.2 Use a Problem Solving Strategy; 5.2 Properties of Exponents and Scientific Notation; 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials

M31-REV-001

I can calculate a percentage, percent increase, or percent decrease.

OpenStax: **Primary:** 2.2 Use a Problem Solving Strategy

SUPPORTING SKILLS

- R Calculate a percentage of a quantity or determine what percent one quantity is of another.
  SK-NUM-PERCENT-CALCULATE · inherited from middle school
- R Calculate and interpret percent increase or decrease.
  SK-NUM-PERCENT-CHANGE · inherited from middle school

M31-REV-002

I can simplify an expression using exponent properties.

OpenStax: **Primary:** 5.2 Properties of Exponents and Scientific Notation

SUPPORTING SKILLS

- R Rewrite a negative exponent using a reciprocal.
  SK-EXP-NEGATIVE · first introduced in Math 21
- R Apply the power rule for exponents.
  SK-EXP-POWER · first introduced in Math 12
- R Apply the product rule for exponents.
  SK-EXP-PRODUCT · first introduced in Math 12
- R Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12
- R Apply the zero-exponent rule.
  SK-EXP-ZERO · first introduced in Math 21

M31-REV-003

I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.

OpenStax: **Primary:** 5.1 Add and Subtract Polynomials

SUPPORTING SKILLS

- R Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M31-REV-004

I can expand a product of two binomials.

OpenStax: **Primary:** 5.3 Multiply Polynomials

SUPPORTING SKILLS

- A Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school
- R Multiply two binomials using distribution.
  SK-POLY-BINOMIAL-MULTIPLY · first introduced in Math 12

Linear Functions and Function Notation REQUIRED

OpenStax coverage: **Primary:** College Algebra 2e 3.1 Functions and Function Notation; College Algebra 2e 3.3 Rates of Change and Behavior of Graphs; College Algebra 2e 3.2 Domain and Range; College Algebra 2e 4.1 Linear Functions; Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables

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M31-LIN-001

I can interpret function notation as a statement about input and output.

OpenStax: **Primary:** 3.1 Functions and Function Notation

SUPPORTING SKILLS

- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Identify the input and output represented by function notation.
  SK-FUNC-NOTATION-READ · first introduced in Math 12
- I Use function notation to express and solve an input-output question.
  SK-FUNC-NOTATION-SOLVE

M31-LIN-002

I can determine whether a graph or table represents a function.

OpenStax: **Primary:** 3.1 Functions and Function Notation

SUPPORTING SKILLS

- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Apply the vertical line test to a graph.
  SK-FUNC-VERTICAL-LINE · first introduced in Math 12

M31-LIN-003

I can evaluate a function or solve for an input using a graph, table, or formula.

OpenStax: **Primary:** 3.1 Functions and Function Notation

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Use function notation to express and solve an input-output question.
  SK-FUNC-NOTATION-SOLVE · first introduced in Math 31
- D Solve for an input associated with a specified output.
  SK-FUNC-SOLVE-INPUT · first introduced in Math 21

M31-LIN-004

I can identify independent and dependent variables in a relationship.

OpenStax: **Primary:** 3.1 Functions and Function Notation

SUPPORTING SKILLS

- I Identify independent and dependent variables.
  SK-FUNC-INDEPENDENT-DEPENDENT
- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12

M31-LIN-005

I can calculate average rate of change from two points.

OpenStax: **Primary:** 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

- D Calculate average rate of change over an interval.
  SK-FUNC-AROC · first introduced in Math 21
- A Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12

M31-LIN-006

I can interpret interval notation and identify increasing or decreasing intervals.

OpenStax: **Primary:** 3.2 Domain and Range

SUPPORTING SKILLS

- I Describe increasing or decreasing behavior over intervals.
  SK-FUNC-INTERVAL-BEHAVIOR
- D Translate between inequality and interval notation.
  SK-NOT-INTERVAL · first introduced in Math 12

M31-LIN-007

I can determine total change from an average rate of change over an interval.

OpenStax: **Primary:** 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

- A Calculate average rate of change over an interval.
  SK-FUNC-AROC · first introduced in Math 21
- I Use average rate of change and interval length to calculate total change.
  SK-FUNC-AROC-TOTAL-CHANGE
- A Add, subtract, multiply, and divide signed integers.
  SK-NUM-INTEGER-OPERATE · inherited from middle school

M31-LIN-008

I can determine and interpret the units of an average rate of change.

OpenStax: **Primary:** 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

- I Determine rate units from input- and output-axis units.
  SK-FUNC-AROC-UNITS
- D Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M31-LIN-009

I can identify a linear function from a table, graph, or formula.

OpenStax: **Primary:** 4.1 Linear Functions

SUPPORTING SKILLS

- I Use constant differences or ratios to distinguish linear and exponential patterns.
  SK-EXP-DIFFERENCE-RATIO
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21
- D Use tabular patterns to predict graph shape and behavior.
  SK-REP-TABLE-GRAPH · first introduced in Math 21

M31-LIN-010

I can interpret the slope and intercept of a linear model.

OpenStax: **Primary:** 4.1 Linear Functions

SUPPORTING SKILLS

- A Plot an intercept accurately on a coordinate plane.
  SK-LIN-PLOT-INTERCEPT · first introduced in Math 12
- A Determine slope from a graph.
  SK-LIN-SLOPE-GRAPH · first introduced in Math 12
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M31-LIN-011

I can determine the intersection of two linear relationships graphically or algebraically.

OpenStax: **Primary:** 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- A Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- A Add equations to eliminate a variable.
  SK-SYS-ELIMINATE · first introduced in Math 12
- A Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12
- A Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE · first introduced in Math 12

M31-LIN-012

I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

OpenStax: **Primary:** 4.1 Linear Functions

SUPPORTING SKILLS

- A Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- A Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT · first introduced in Math 12
- D Recognize that distinct parallel lines have equal slopes.
  SK-LIN-PARALLEL-SLOPE · first introduced in Math 12
- D Determine the negative reciprocal of a nonzero slope.
  SK-LIN-PERPENDICULAR-SLOPE · first introduced in Math 12

Function Behavior and Structure REQUIRED

OpenStax coverage: **Primary:** 3.1 Functions and Function Notation; 3.2 Domain and Range; 3.5 Transformation of Functions; 3.4 Composition of Functions; 3.7 Inverse Functions; 3.3 Rates of Change and Behavior of Graphs

M31-FUN-001

I can determine whether a relation is a function.

OpenStax: **Primary:** 3.1 Functions and Function Notation

SUPPORTING SKILLS

- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Apply the vertical line test to a graph.
  SK-FUNC-VERTICAL-LINE · first introduced in Math 12

M31-FUN-002

I can evaluate a function from a formula, graph, or table.

OpenStax: **Primary:** 3.1 Functions and Function Notation

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Solve for an input associated with a specified output.
  SK-FUNC-SOLVE-INPUT · first introduced in Math 21

M31-FUN-003

I can determine domain and range from a formula, graph, or table.

OpenStax: **Primary:** 3.2 Domain and Range

SUPPORTING SKILLS

- I Determine domain restrictions from denominators, even roots, and logarithms.
  SK-FUNC-DOMAIN-FORMULA
- D Read domain and range from a graph.
  SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12
- D Read domain and range from a table.
  SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12

M31-FUN-004

I can determine algebraic domain restrictions caused by denominators and even roots.

OpenStax: **Primary:** 3.2 Domain and Range

SUPPORTING SKILLS

- A Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D Require an even-root radicand to be nonnegative when determining domain.
  SK-FUNC-DOMAIN-EVEN-ROOT · first introduced in Math 21
- D Determine domain restrictions from denominators, even roots, and logarithms.
  SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- A Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS · first introduced in Math 21

M31-FUN-005

I can graph a piecewise function from its formula.

OpenStax: **Primary:** 3.2 Domain and Range

SUPPORTING SKILLS

- A Describe increasing or decreasing behavior over intervals.
  SK-FUNC-INTERVAL-BEHAVIOR · first introduced in Math 31
- I Select the correct rule of a piecewise function for a given input.
  SK-FUNC-PIECEWISE-EVALUATE
- I Graph each rule on its stated domain with correct endpoints.
  SK-FUNC-PIECEWISE-GRAPH

M31-FUN-006

I can write a piecewise formula that represents a graph.

OpenStax: **Primary:** 3.2 Domain and Range

SUPPORTING SKILLS

- A Describe increasing or decreasing behavior over intervals.
  SK-FUNC-INTERVAL-BEHAVIOR · first introduced in Math 31
- I Translate a segmented graph into a piecewise formula.
  SK-FUNC-PIECEWISE-WRITE
- A Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT · first introduced in Math 12

M31-FUN-007

I can graph a transformed function using shifts, stretches, compressions, and reflections.

OpenStax: **Primary:** 3.5 Transformation of Functions

SUPPORTING SKILLS

- I Interpret transformations applied to function inputs.
  SK-FUNC-TRANSFORM-HORIZONTAL
- I Distinguish stretches from compressions and horizontal effects from vertical effects.
  SK-FUNC-TRANSFORM-SCALE
- I Interpret transformations applied to function outputs.
  SK-FUNC-TRANSFORM-VERTICAL
- A Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-FUN-008

I can describe how a transformation affects domain, range, or asymptotes.

OpenStax: **Primary:** 3.5 Transformation of Functions

SUPPORTING SKILLS

- I Exchange domain and range between a function and its inverse.
  SK-FUNC-INVERSE-DOMAIN-RANGE
- D Interpret transformations applied to function inputs.
  SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- D Distinguish stretches from compressions and horizontal effects from vertical effects.
  SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- D Interpret transformations applied to function outputs.
  SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31

M31-FUN-009

I can evaluate a composition from formulas, graphs, or tables.

OpenStax: **Primary:** 3.4 Composition of Functions

SUPPORTING SKILLS

- I Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA
- I Follow intermediate values to compose functions from tables.
  SK-FUNC-COMPOSE-TABLE
- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I Read and write notation for function composition and inverse functions.
  SK-FUNC-STRUCTURE-NOTATION

M31-FUN-010

I can write a formula for a composition of two functions.

OpenStax: **Primary:** 3.4 Composition of Functions

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- D Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31

M31-FUN-011

I can find an inverse function algebraically.

OpenStax: **Primary:** 3.7 Inverse Functions

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- I Exchange variables and solve to obtain an inverse formula.
  SK-FUNC-INVERSE-ALGEBRA
- I Exchange input and output values when interpreting an inverse.
  SK-FUNC-INVERSE-SWAP
- A Read and write notation for function composition and inverse functions.
  SK-FUNC-STRUCTURE-NOTATION · first introduced in Math 31

M31-FUN-012

I can evaluate and interpret an inverse from a graph, formula, or table.

OpenStax: **Primary:** 3.7 Inverse Functions

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Exchange variables and solve to obtain an inverse formula.
  SK-FUNC-INVERSE-ALGEBRA · first introduced in Math 31
- D Exchange input and output values when interpreting an inverse.
  SK-FUNC-INVERSE-SWAP · first introduced in Math 31

M31-FUN-013

I can determine and interpret the units of an inverse function.

OpenStax: **Primary:** 3.7 Inverse Functions

SUPPORTING SKILLS

- I Reverse input and output units when interpreting an inverse.
  SK-FUNC-INVERSE-UNITS
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M31-FUN-014

I can explain how the domain and range of a function relate to those of its inverse.

OpenStax: **Primary:** 3.7 Inverse Functions

SUPPORTING SKILLS

- I Exchange domain and range between a function and its inverse.
  SK-FUNC-INVERSE-DOMAIN-RANGE
- A Exchange input and output values when interpreting an inverse.
  SK-FUNC-INVERSE-SWAP · first introduced in Math 31

M31-FUN-015

I can determine the range of a function by analyzing the domain of its inverse.

OpenStax: **Primary:** 3.7 Inverse Functions

SUPPORTING SKILLS

- A Determine domain restrictions from denominators, even roots, and logarithms.
  SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- D Exchange domain and range between a function and its inverse.
  SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31

M31-FUN-016

I can verify algebraically that two functions are inverses.

OpenStax: **Primary:** 3.7 Inverse Functions

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- A Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- I Verify inverse functions using composition.
  SK-FUNC-INVERSE-VERIFY

M31-FUN-017

I can interpret the meaning and units of a composition.

OpenStax: **Primary:** 3.4 Composition of Functions

SUPPORTING SKILLS

- D Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D Follow intermediate values to compose functions from tables.
  SK-FUNC-COMPOSE-TABLE · first introduced in Math 31
- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- D Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M31-FUN-018

I can classify a graph or table as concave up or concave down using changes in average rate of change.

OpenStax: **Primary:** 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

- A Calculate average rate of change over an interval.
  SK-FUNC-AROC · first introduced in Math 21
- I Use successive average rates of change to classify concavity.
  SK-FUNC-CONCAVITY-AROC
- D Use tabular patterns to predict graph shape and behavior.
  SK-REP-TABLE-GRAPH · first introduced in Math 21

M31-FUN-019

I can determine whether a function is invertible using the horizontal line test.

OpenStax: **Primary:** 3.7 Inverse Functions

SUPPORTING SKILLS

- I Apply the horizontal line test for invertibility.
  SK-FUNC-INVERSE-HLT
- I Test whether a function is one-to-one on a domain.
  SK-FUNC-ONE-TO-ONE

M31-FUN-020

I can graph an inverse function as a reflection across the line `y = x`.

OpenStax: **Primary:** 3.7 Inverse Functions

SUPPORTING SKILLS

- A Exchange domain and range between a function and its inverse.
  SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31
- D Exchange input and output values when interpreting an inverse.
  SK-FUNC-INVERSE-SWAP · first introduced in Math 31
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

Quadratic Functions and Equations

REQUIRED

OpenStax coverage: **Primary:** 5.1 Quadratic Functions; 2.5 Quadratic Equations
Supplemental: 5.3 Graphs of Polynomial Functions; 5.5 Zeros of Polynomial Functions

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M31-QUAD-001

I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.

OpenStax: **Primary:** 5.1 Quadratic Functions

SUPPORTING SKILLS

- A Use successive average rates of change to classify concavity.
  SK-FUNC-CONCAVITY-AROC · first introduced in Math 31
- A Identify standard, factored, and vertex forms.
  SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- D Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS · first introduced in Math 21
- D Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX · first introduced in Math 21
- R Recall and correctly use quadratic-function terms including parabola, vertex, axis of symmetry, intercept, zero, and solution.
  SK-QUAD-VOCABULARY · first introduced in Math 21

M31-QUAD-002

I can connect standard, factored, and vertex forms to features of a quadratic graph.

OpenStax: **Primary:** 5.1 Quadratic Functions
Supplemental: 5.3 Graphs of Polynomial Functions

SUPPORTING SKILLS

- D Identify standard, factored, and vertex forms.
  SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- A Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS · first introduced in Math 21
- A Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX · first introduced in Math 21
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-QUAD-003

I can rewrite a quadratic function in a form that reveals a requested feature.

OpenStax: **Primary:** 5.1 Quadratic Functions

SUPPORTING SKILLS

- A Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D Create a perfect-square trinomial by completing the square.
  SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21
- D Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
  SK-QUAD-FORM-CONVERT · first introduced in Math 21

M31-QUAD-004

I can solve a quadratic equation by factoring or using the quadratic formula.

OpenStax: **Primary:** 2.5 Quadratic Equations

SUPPORTING SKILLS

- A Apply the zero-product property.
  SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- A Use the discriminant to predict the number and type of solutions.
  SK-QUAD-DISCRIMINANT · first introduced in Math 21
- A Substitute coefficients accurately into the quadratic formula.
  SK-QUAD-FORMULA · first introduced in Math 21
- D Select an efficient quadratic-solving method from equation structure.
  SK-QUAD-METHOD-SELECT · first introduced in Math 21

M31-QUAD-005

I can graph a quadratic function using its algebraic structure and key features.

OpenStax: **Primary:** 5.1 Quadratic Functions

SUPPORTING SKILLS

- A Identify standard, factored, and vertex forms.
  SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- D Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS · first introduced in Math 21
- D Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX · first introduced in Math 21
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-QUAD-006

I can construct a quadratic function from specified zeros, a vertex, or graphical information.

OpenStax: **Primary:** 5.1 Quadratic Functions
Supplemental: 5.5 Zeros of Polynomial Functions

SUPPORTING SKILLS

- A Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- I Build a polynomial from zeros, multiplicities, and a point.
  SK-POLY-CONSTRUCT
- A Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
  SK-QUAD-FORM-CONVERT · first introduced in Math 21

M31-QUAD-007

I can create and interpret a quadratic trajectory model.

OpenStax: **Primary:** 5.1 Quadratic Functions

SUPPORTING SKILLS

- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- D Interpret vertex and zeros in an application.
  SK-QUAD-MODEL-FEATURES · first introduced in Math 21
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

Exponential Functions and Models

REQUIRED

OpenStax coverage: **Primary:** 6.1 Exponential Functions; 6.2 Graphs of Exponential Functions

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M31-EXP-001

I can distinguish linear change from exponential change using differences, ratios, or percent change.

OpenStax: **Primary:** 6.1 Exponential Functions

SUPPORTING SKILLS

- D Use constant differences or ratios to distinguish linear and exponential patterns.
  SK-EXP-DIFFERENCE-RATIO · first introduced in Math 31
- D Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21
- A Calculate and interpret percent increase or decrease.
  SK-NUM-PERCENT-CHANGE · inherited from middle school

M31-EXP-002

I can identify initial value and growth or decay factor in an exponential model.

OpenStax: **Primary:** 6.1 Exponential Functions

SUPPORTING SKILLS

- I Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT
- I Convert a percent increase or decrease to a growth or decay factor.
  SK-EXP-PERCENT-FACTOR
- I Recall and correctly use exponential-function terms including initial value, growth or decay factor, growth or decay rate, and asymptote.
  SK-EXP-VOCABULARY
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

M31-EXP-003

I can write an exponential function from a table, graph, or context.

OpenStax: **Primary:** 6.1 Exponential Functions

SUPPORTING SKILLS

- D Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- I Determine an exponential model from tabular values.
  SK-EXP-MODEL-TABLE
- I Solve for an exponential model using two points.
  SK-EXP-MODEL-TWO-POINTS
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-EXP-004

I can evaluate and interpret an exponential model.

OpenStax: **Primary:** 6.1 Exponential Functions

SUPPORTING SKILLS

- D Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M31-EXP-005

I can calculate growth under periodic compounding.

OpenStax: **Primary:** 6.1 Exponential Functions

SUPPORTING SKILLS

- I Substitute correctly into a periodic-compounding model.
  SK-EXP-COMPOUND-PERIODIC
- A Convert a percent increase or decrease to a growth or decay factor.
  SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- A Calculate a percentage of a quantity or determine what percent one quantity is of another.
  SK-NUM-PERCENT-CALCULATE · inherited from middle school

M31-EXP-006

I can calculate growth under continuous compounding.

OpenStax: **Primary:** 6.1 Exponential Functions

SUPPORTING SKILLS

- I Substitute correctly into a continuous-compounding model.
  SK-EXP-COMPOUND-CONTINUOUS
- A Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31

M31-EXP-007

I can determine an exponential formula from two points or other sufficient data.

OpenStax: **Primary:** 6.1 Exponential Functions

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D Solve for an exponential model using two points.
  SK-EXP-MODEL-TWO-POINTS · first introduced in Math 31
- A Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL · first introduced in Math 21

M31-EXP-008

I can determine domain, range, intercepts, and asymptotes of an exponential function.

OpenStax: **Primary:** 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

- I Identify horizontal asymptotic behavior of an exponential graph.
  SK-EXP-ASYMPTOTE
- D Determine domain restrictions from denominators, even roots, and logarithms.
  SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-EXP-009

I can compare the growth rates of two exponential functions.

OpenStax: **Primary:** 6.1 Exponential Functions

SUPPORTING SKILLS

- I Compare exponential growth rates expressed in different forms.
  SK-EXP-COMPARE-GROWTH
- D Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21

M31-EXP-010

I can determine the percent growth or decay represented by a continuous exponential model.

OpenStax: **Primary:** 6.1 Exponential Functions

SUPPORTING SKILLS

- I Convert between equivalent discrete- and continuous-growth forms.
  SK-EXP-DISCRETE-CONTINUOUS
- D Convert a percent increase or decrease to a growth or decay factor.
  SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- A Calculate and interpret percent increase or decrease.
  SK-NUM-PERCENT-CHANGE · inherited from middle school

Logarithmic Functions and Models REQUIRED

OpenStax coverage: **Primary:** 6.3 Logarithmic Functions; 6.5 Logarithmic Properties; 6.6 Exponential and Logarithmic Equations; 6.7 Exponential and Logarithmic Models; 6.4 Graphs of Logarithmic Functions
Supplemental: 6.2 Graphs of Exponential Functions

M31-LOG-001

I can convert between logarithmic and exponential forms.

OpenStax: **Primary:** 6.3 Logarithmic Functions

SUPPORTING SKILLS

- I Convert between logarithmic and exponential forms.
  SK-LOG-CONVERT
- I Interpret a logarithm as an exponent.
  SK-LOG-DEFINITION
- I Read logarithmic notation and identify the base, argument, and corresponding exponential statement.
  SK-LOG-NOTATION-READ

M31-LOG-002

I can evaluate a logarithm using its exponential meaning.

OpenStax: **Primary:** 6.3 Logarithmic Functions

SUPPORTING SKILLS

- I Rewrite exponential expressions using a common base when possible.
  SK-EXP-REWRITE-BASE
- D Interpret a logarithm as an exponent.
  SK-LOG-DEFINITION · first introduced in Math 31

M31-LOG-003

I can expand or condense an expression using product, quotient, and power properties.

OpenStax: **Primary:** 6.5 Logarithmic Properties

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Apply the logarithm power property in either direction.
  SK-LOG-POWER
- I Apply the logarithm product property in either direction.
  SK-LOG-PRODUCT
- I Apply the logarithm quotient property in either direction.
  SK-LOG-QUOTIENT

M31-LOG-004

I can solve an exponential equation using logarithms.

OpenStax: **Primary:** 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- A Apply the logarithm power property in either direction.
  SK-LOG-POWER · first introduced in Math 31
- I Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP

M31-LOG-005

I can solve a logarithmic equation using exponential form and check domain restrictions.

OpenStax: **Primary:** 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

- A Convert between logarithmic and exponential forms.
  SK-LOG-CONVERT · first introduced in Math 31
- I Reject solutions that make a logarithm argument nonpositive.
  SK-LOG-DOMAIN-CHECK
- I Combine or isolate logarithms before converting to exponential form.
  SK-LOG-SOLVE-LOG

M31-LOG-006

I can calculate and interpret doubling time or half-life.

OpenStax: **Primary:** 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- I Translate a doubling-time or half-life question into an exponential equation.
  SK-LOG-DOUBLING-HALF
- A Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M31-LOG-007

I can determine domain, range, intercepts, and asymptotes of a logarithmic function.

OpenStax: **Primary:** 6.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

- D Determine domain restrictions from denominators, even roots, and logarithms.
  SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- I Identify vertical asymptotic behavior of a logarithmic graph.
  SK-LOG-ASYMPTOTE
- A Reject solutions that make a logarithm argument nonpositive.
  SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-LOG-008

I can explain how exponential and logarithmic functions undo one another.

OpenStax: **Primary:** 6.3 Logarithmic Functions

SUPPORTING SKILLS

- A Exchange input and output values when interpreting an inverse.
  SK-FUNC-INVERSE-SWAP · first introduced in Math 31
- A Verify inverse functions using composition.
  SK-FUNC-INVERSE-VERIFY · first introduced in Math 31
- D Interpret a logarithm as an exponent.
  SK-LOG-DEFINITION · first introduced in Math 31

M31-LOG-009

I can solve a multi-step exponential or logarithmic equation.

OpenStax: **Primary:** 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

- A Reject solutions that make a logarithm argument nonpositive.
  SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- A Apply the logarithm power property in either direction.
  SK-LOG-POWER · first introduced in Math 31
- A Apply the logarithm product property in either direction.
  SK-LOG-PRODUCT · first introduced in Math 31
- A Apply the logarithm quotient property in either direction.
  SK-LOG-QUOTIENT · first introduced in Math 31
- D Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31
- D Combine or isolate logarithms before converting to exponential form.
  SK-LOG-SOLVE-LOG · first introduced in Math 31

M31-LOG-010

I can determine an intersection of exponential functions graphically or algebraically.

OpenStax: **Primary:** 6.6 Exponential and Logarithmic Equations
Supplemental: 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

- A Compare exponential growth rates expressed in different forms.
  SK-EXP-COMPARE-GROWTH · first introduced in Math 31
- D Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31
- A Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- A Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M31-LOG-011

I can rewrite a discrete exponential model as an equivalent continuous exponential model.

OpenStax: **Primary:** 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- D Convert between equivalent discrete- and continuous-growth forms.
  SK-EXP-DISCRETE-CONTINUOUS · first introduced in Math 31
- A Convert a percent increase or decrease to a growth or decay factor.
  SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- A Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31

M31-LOG-012

I can solve and interpret an exponential or logarithmic modeling problem.

OpenStax: **Primary:** 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- D Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- A Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31
- A Combine or isolate logarithms before converting to exponential form.
  SK-LOG-SOLVE-LOG · first introduced in Math 31
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

Extension Content

EXTENSION

OpenStax coverage: **Primary:** 3.6 Absolute Value Functions; 3.5 Transformation of Functions; 6.7 Exponential and Logarithmic Models

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M31-EXT-001

I can analyze an absolute-value function using transformations.

OpenStax: **Primary:** 3.6 Absolute Value Functions

SUPPORTING SKILLS

- A Interpret transformations applied to function inputs.
  SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- A Distinguish stretches from compressions and horizontal effects from vertical effects.
  SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- A Interpret transformations applied to function outputs.
  SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31
- A Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-EXT-002

I can determine even or odd symmetry from a graph or formula.

OpenStax: **Primary:** 3.5 Transformation of Functions

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Test a formula or graph for even or odd symmetry.
  SK-FUNC-SYMMETRY

M31-EXT-003

I can interpret a contextual logarithmic scale.

OpenStax: **Primary:** 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- A Interpret a logarithm as an exponent.
  SK-LOG-DEFINITION · first introduced in Math 31
- I Interpret equal intervals on a logarithmic scale as equal multiplicative changes.
  SK-LOG-SCALE-INTERPRET
- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21